December 3, 2019

# FOURIER COEFFICIENTS OF AUTOMORPHIC FORMS

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Is there a Fourier expansion for automorphic forms? Do Fourier coefficients characterize automorphic forms?

Whittaker coefficients are just an integral over unipotent radical of Borel against some generic character. Similarly, there is a more general theory of Fourier coefficients for automorphic forms which are associated with **unipotent orbits**.

## 1. Generalities of unipotent orbits

Although I will talk about **nilpotent orbits** in **semisimple** Lie **algebra**, nothing much is lost... We follow [ColMcG].

Given a complex semisimple Lie algebra $\mathfrak{g}$, an element $X \in \mathfrak{g}$ is **nilpotent** (**semisimple**, resp.) if $[X, -]$ is a nilpotent (semisimple, resp.) operator on $\mathfrak{g}$. The Jordan decomposition says that for general $X \in \mathfrak{g}$ there is a unique decomposition $X = X_s + X_n$ where $X_s$ is semisimple, $X_n$ is nilpotent and $[X_s, X_n] = 0$.

To define semisimplicity and nilpotence for coadjoint orbits (which might be neat because coadjoint orbits have extra structure of being symplectic), we need to do more as $\mathfrak{g}^*$ is a priori not a Lie algebra. We can do two things which turn out to be equivalent.

- Lazy definition: use Killing form to realize $\mathfrak{g} \cong \mathfrak{g}^*$.
- Sophiscated definition: $X \in \mathfrak{g}^*$ is semisimple if
$$\mathfrak{g}^X = \{ Y \in \mathfrak{g} \mid \mathrm{ad}^*_Y X = 0 \},$$
is reductive; $X$ is nilpotent if $X \mid \mathfrak{g}^X \equiv 0$.

### 1.1. Semisimple orbits.

For semisimple $X \in \mathfrak{g}$, the reductive centralizer $\mathfrak{g}^X$ has a Cartan subalgebra $\mathfrak{h}$ containing $X$. Moreover, any Cartan subalgebra $\mathfrak{g} \supset \mathfrak{h}$ is the centralizer of a regular semisimple element (regular means centralizer is a torus). Also the set of regular semisimple elements is covered by conjugations of a Cartan. The **rank** of $\mathfrak{g}$ is the exponent of the largest power of $T$ dividing the characteristic polynomial of $\mathrm{ad}_X$ for a regular semisimple $X$. In turn, any semisimple element can be conjugated into a fixed Cartan. More precisely, given a Cartan $\mathfrak{h} \subset \mathfrak{g}$,
$$\left\{ \begin{array}{c} \text{semisimple} \\ \text{orbits} \end{array} \right\} = \mathfrak{h}/W$$

In particular there are **infinitely** many semisimple orbits.

- An adjoint orbit is semisimple iff it is closed in $\mathfrak{g}$.
- $X \in \mathfrak{g}$ is semisimple, then $G_{sc}^X$ is connected.
- A semisimple adjoint orbit is simply connected.

So we don't expect much from the topology of semisimple orbits. No closure relations, no fundamental group.

1.2. **Nilpotent orbits.** What about nilpotent orbits? Unlike semisimple orbits there are only **finitely many** nilpotent orbits. We derive the Dynkin-Kostant classification.

- It is illuminating to do $\mathfrak{sl}_n$ first. We see that nilpotent orbits correspond to partitions of $n$, which sends a nilpotent element to the partitions determined by the block sizes in its Jordan normal form. One has similar partition-type classifications for Lie algebras of type $A, B, C, D$:
  - ($A_n$) Nilpotent orbits in $\mathfrak{sl}_n$ = partitions of $n$.
  - ($B_n$) Nilpotent orbits in $\mathfrak{so}_{2n+1}$ = partitions of $2n+1$, in which even parts occur with even multiplicity.
  - ($C_n$) Nilpotent orbits in $\mathfrak{sp}_{2n}$ = partitions of 2n, in which odd parts occur with even multiplicity.
  - ($D_n$) Nilpotent orbits in $\mathfrak{so}_{2n}$ = partitions of $2n$, in which even parts occur with even multiplcity, **except that "very even" partitions (=all entries are even, and each has even multiplicity) correspond to two orbits**. But you know $D$ is always a little weird so it's okay.
- NB: The unipotent orbit corresponding to the superdiagonal of the parabolic subgroup of $\mathrm{GL}_n$ associated to the partition $\mathbf{a}$ actually corresponds to, via the partition-type classification, $\mathbf{a}^t$, the transpose of $\mathbf{a}$ (as a Young diagram)!
- An elt is nilpotent iff it is in nilradical of some Borel. This suggests that one could always complete a nilpotent element into $\mathfrak{sl}_2$ in $\mathfrak{g}$. So nilpotent orbits = conjugacy classes of nonzero homomorphisms $\mathfrak{sl}_2 \longrightarrow \mathfrak{g}$. This then turn into classifying semisimple elements arising as $H$ in $\mathfrak{sl}_2$ (usual $HXY$ notation). Finally one sees that such $H$ always satisfies $\alpha(H) \in \{0, 1, 2\}$ for $\alpha \in \Delta$, making the set of these "distinguished semisimple elements" a finite set (3 choices for each vertex in the Dynkin diagram). It is tricky to see which labeling is a distinguished element.
- More generally let $\mathfrak{g} = \oplus_{i \in \mathbb{Z}} \mathfrak{g}_i$ be the $\mathrm{ad}_H$-eigenspace decomposition of $\mathfrak{g}$. Then, $\mathcal{O}_X$, the adjoint orbit of $X$, has $\dim(\mathcal{O}_X) = \dim \mathfrak{g} - \dim \mathfrak{g}_0 - \dim \mathfrak{g}_1$.

We now know at least there are finitely many nilpotent orbits. We order it by closure-inclusion relation. Obviously this is a partial order (no two different orbits have the same Zariski closure, which follows from that an orbit is dense in its closure) and the structure varies from group to group, but there are also some parts that are in common for all Lie algebras.

- Obviously 0 is the minimum. You can see that 0 is contained in any orbit closure by reducing to $\mathfrak{sl}_2$.
- The first nontrivial result is that there is also a **maximum**, called the **principal orbit** or the **regular orbit**! This corresponds to $X = \sum_{\alpha \in \Delta} X_\alpha$, and the corresponding $H$ is such that $\alpha(H) = 2$ for all $\alpha \in \Delta$. It is even (i.e. $\mathfrak{g}_{\mathrm{odd}} = 0$), dense open in $\mathcal{N}$, the set of nilpotent elements, and is of dimension $\dim \mathfrak{g} - \operatorname{rank} \mathfrak{g}$. Thus the corresponding weighted Dynkin diagram has every node labeled 2.
- There is then the **unique second largest nilpotent orbit**! It is called the **subregular orbit**. We say $X \in \mathfrak{g}$ is **subregular** if $\dim \mathfrak{g}^X = \operatorname{rank} \mathfrak{g} + 2$. What we claim here is thus that subregular nilpotent elements are all conjugate, and these are open dense in the boundary of the principal orbit (and of dimension $\dim \mathfrak{g} - \operatorname{rank} \mathfrak{g} - 2$). Although there is uniqueness, at least to me the nature of the subregular orbit is pretty mysterious (case by case calculation after reducing to rank 2 cases).

- There is also the **unique second smallest orbit**! It is called the **minimal orbit**, and it is the orbit of any nonzero highest weight vector. The dimension of the orbit is

$$1 + \#\{\alpha \in \Delta \mid \alpha \text{ not orthogonal to the chosen highest weight}\}.$$

- We note that for partition-type classifications the order is just easy, namely $(r_1 \geq r_2 \geq \cdots \geq r_m \geq 0) \geq (l_1 \geq l_2 \geq \cdots \geq l_m \geq 0)$ iff $r_1 + \cdots + r_i \geq l_1 + \cdots + l_i$ for all $1 \leq i \leq m$.

We end by briefly reviewing Bala-Carter theory which gives a way to classify all nilpotent orbits from Dynkin diagram data.

- The main theorem in this regard is the following: any two minimal Levi subalgebras containing a fixed nilpotent $X \in \mathfrak{g}$ are $G_{\mathrm{ad}}^X$-conjugate. We say $X$ is **distinguished** if the only Levi containing $X$ is $\mathfrak{g}$. This turns out to be equivalent to $\dim \mathfrak{g}_0 = \dim \mathfrak{g}_2$, and if so the distinguished orbit becomes **even**. The study of adjoint orbits can be done by studying distinguished adjoint orbits for all Levis of $\mathfrak{g}$, so subsequently all $[\mathfrak{l}, \mathfrak{l}]$ for all Levis $\mathfrak{l} \subset \mathfrak{g}$.
- The $\mathfrak{sl}_2$-theory attaches to $\mathcal{O}_X$ a parabolic subalgebra $\mathfrak{q} \subset \mathfrak{g}$. This sees the distinguishedness of $X$, as $\mathcal{O}_X$ is distinguished iff $\mathfrak{q} = \mathfrak{l} \oplus \mathfrak{u}$ (Levi decomposition) satisfies $\dim \mathfrak{l} = \dim \mathfrak{u}/[\mathfrak{u}, \mathfrak{u}]$. Conversely, any parabolic subalgebra with that dimension condition arises as the associated parabolic subalgebra to a distinguished adjoint orbit. This gives a one-to-one correspondence

$$\left\{ \text{ Nilpotent orbits of } \mathfrak{g} \right\} = \left\{ \begin{array}{c} G_{\mathrm{ad}}\text{-conjugacy classes of pairs } (\mathfrak{l}, \mathfrak{p}_{\mathfrak{l}}) \text{ where } \mathfrak{l} \text{ is} \\ \text{Levi in } \mathfrak{g} \text{ and } \mathfrak{p}_\ell \text{ is a distinguished parabolic} \\ \text{subalgebra of } [\mathfrak{l}, \mathfrak{l}] \end{array} \right\}.$$

## 2. Local theory

I will just try to translate Moeglin-Waldspurger. For a local representation, any representation $\pi$

## 3. Fourier coefficients and unipotent orbits

Now we describe [Ginzburg]. We work with split reductive classical group $G$ over a number field $F$ (???dubious still). This gives a partition-type classification of unipotent orbits. To be more precise, given a partition $\mathbf{a} = (m_1 \cdots m_p)$ with $m_1 \geq \cdots \geq m_p > 0$, the corresponding cocharacter $h_{\mathbf{a}}(t)$ is given by first associating to each $m_i$ $\mathrm{diag}(t^{m_i-1}, t^{m_i-3}, \cdots, t^{-(m_i-3)}, t^{-(m_i-1)})$ and then rearranging everything in decreasing order of $t^{k_i}$. Let

- $L$ be the subgroup of $G$ generated by all roots in $\mathfrak{g}_{\pm 1}$ (following the previous section's terminology),
- $V$ be the unipotent subgroup of $L$ generated by the roots in $\mathfrak{g}_2$.

Now a **Fourier coefficient** attached to a unipotent orbit actually needs some auxiliary data, so that there might be multiple (even infinitely many!) Fourier coefficients associated to a single unipotent orbit.

**Definition 3.1** (Fourier coefficient)**.** *Let* $\mathbf{a}$ *be a unipotent orbit. The additional datum is that we pick* $C$*, the stabilizer in* $L$ *of a representative of the unipotent orbit of* $\mathbf{a}$*. We know* $C$ *is reductive but can in general vary as we will see below. Let* $\psi_V : V^{\mathrm{ab}}(F\backslash \mathbb{A}_F) \longrightarrow \mathbb{C}$ *be an additive character such that the stabilizer of* $\psi_V$ *inside* $L(F)$*, denoted as* $C_1$*, is a form of* $C$*. We extend* $\psi_V$ *trivially to the group* $V(F)\backslash V(\mathbb{A}_F)$*; then*

$$\mathcal{F}_{\mathbf{a}}(\varphi)(g) = \int_{V(F)\backslash V(\mathbb{A})} \varphi(vg)\psi_V(v)dv,$$

*is a Fourier coefficient associated with* $\mathbf{a}$*.*

**Example 3.1.** If $G = \mathrm{Sp}_4$ and $\mathbf{a} = (2^2)$, then $h_{\mathbf{a}}(t) = \mathrm{diag}(t, t, t^{-1}, t^{-1})$, $L = \mathrm{GL}_2$, and $V$ is of form

$$\begin{pmatrix} 1 & & x & y \\ & 1 & z & x \\ & & 1 & \\ & & & 1 \end{pmatrix}.$$

Any stabilizer of a representative of the unipotent orbit is a one-dimensional torus. We can choose $\psi_V$ as $\psi_V \begin{pmatrix} 1 & & x & y \\ & 1 & z & x \\ & & 1 & \\ & & & 1 \end{pmatrix} = \psi(\beta y + \gamma z)$ for an additive character $\psi$ and $\beta, \gamma \in F^\times$. The stabilizer of $\psi_V$ in $L$ can have different forms depending on whether $-\beta\gamma$ is a square or not. Namely, if $-\beta\gamma$ is a square, then $C_1(F) \cong \mathrm{GL}_1(F)$, whereas if not, $C_1(F)$ is anisotropic.

**Definition 3.2.** *For an automorphic representation $\pi$ of $G(\mathbb{A})$, $\mathcal{O}_G(\pi)$ is the set of unipotent orbits* **a** *which are maximal among unipotent orbits having some nonzero Fourier coefficient.*

**Example 3.2.** For $G = \mathrm{GL}_n$ and $\pi$ cuspidal, we know $\pi$ is generic, which just tautologically means that the unipotent orbit $(n)$ has nonzero Fourier coefficient. As $(n)$ is greater than any other unipotent orbits, $\mathcal{O}_G(\pi) = \{(n)\}$.

In general $\mathcal{O}_G(\pi)$ is nonempty well-defined unless $\pi$ is a trivial representation, as the minimal orbit has nonzero Fourier coefficients (just Fourier coefficients of usual $\mathrm{SL}_2$ case by restricting to the root group generated by the highest root).

**Definition 3.3.** *A partition of certain type is* ***special*** *if it is transpose of some other partition, where transpose here is in a sense of Young diagrams.*

**Theorem 3.1** (Ginzburg-Rallis-Soudry, Moeglin)**.** *Any unipotent orbit in $\mathcal{O}_G(\pi)$ is special.*

*Proof.* We include a vague sketch of proof because it uses a structure that is quite illuminating. Let $U_{\geq j}$ be the group generated by roots that have inner product with $h_{\mathbf{a}}(t) \geq j$ (for example, $V = U_{\geq 2}$). Then, $U_{\geq 1}/[U_{\geq 2}, U_{\geq 2}]$ has the structure of **generalized Heisenberg group**, via $[,] : U_{\geq 1}/U_{\geq 2} \times U_{\geq 1}/U_{\geq 2} \longrightarrow U_{\geq 2}^{\mathrm{ab}}$. This means that there is a natural map $\sigma : U_{\geq 1}/[U_{\geq 2}, U_{\geq 2}] \longrightarrow H_{2m+1}(\mathbb{A})$ where $m$ is the number of simple roots $\alpha$ that satisfy $\langle \alpha, h_{\mathcal{O}} \rangle = 1$ and $H_{2m+1}$ is the usual Heisenberg group, i.e. the group

$$\left\{ \begin{pmatrix} 1 & * & * \\ & I_m & * \\ & & 1 \end{pmatrix} \right\},$$

such that there is a compatible embedding $C \hookrightarrow \mathrm{Sp}_{2m}$, where compatible means the group action of $C$ on $U_{\geq 1}/[U_{\geq 2}, U_{\geq 2}]$ is compatible with the group action of $\mathrm{Sp}_{2m}$ on $H_{2m+1}$ via the two maps.

A great advantage of this viewpoint is that we can use a theta representation of $\widetilde{\mathrm{Sp}}_{2m}(\mathbb{A})$. Let $\theta_\phi^\psi$ be the theta representation such that $\theta_\phi^\psi(\sigma(z)h) = \psi_V(z)\theta_\phi^\psi(h)$ for $z \in U_{\geq 1}/[U_{\geq 2}, U_{\geq 2}]$ with $\sigma(z)$ in the center (i.e. upper right corner of the matrix representation). Then, almost by case-by-case analysis, Nevis proved that for $G$ classical and **a** non-special,

$$f(h) = \int_{U_{\geq 1}(F)\backslash U_{\geq 1}(\mathbb{A})} \theta_\phi^\psi(\sigma(v)h)\varphi(vh)\psi(\mathrm{proj}\,\sigma(v))dv,$$

where proj is the projection to the center,defines an honest function on $\widetilde{C}(\mathbb{A})$. Thus, for a non-special **a** with nontrivial Fourier coefficient, restricting $f$ to $\widetilde{\mathrm{SL}}_2$ formed by highest/lowest roots of $C$ gives a nonconstant function, which means that some nonconstant Fourier coefficient of $\widetilde{\mathrm{SL}}_2(\mathbb{A})$-function is nonzero. That Fourier coefficient is associated with a unipotent orbit that is strictly greater than **a**! □

### 3.1. **Expectations.**

3.1.1. *Cusp forms.* From $p$-adic results of Moeglin-Waldspurger (degenerate Whittaker models...) we expect the following for cuspidal representations.

**Conjecture 3.1.** *If $\pi$ is cuspidal and $\mathbf{a} \in \mathcal{O}_G(\pi)$ is a unipotent orbit, then* **a** *has a representative defined over $F$ that cannot be $G(F)$-conjugated into a proper Levi.*

This greatly restricts partitions that can arise as elements of $\mathcal{O}_G(\pi)$. For example, this is consistent with $G = \mathrm{GL}_n$ case as $(n)$ is the only partition that cannot be conjugated into a proper Levi. On the other hand, if $G = \mathrm{Sp}_{2n}$, then the partitions that cannot be conjugated into a proper Levi are partitions consisted only of even numbers.

3.1.2. *Eisenstein series for* $\mathrm{GL}_n$. We first have an uber-conjecture for automorphic representations of $G = \mathrm{GL}_n$.

**Conjecture 3.2.** *For an automorphic representation $\pi$ of $\mathrm{GL}_n$, $\mathcal{O}_G(\pi)$ is a singleton.*

Let us illustrate finer conjectures.

**Conjecture 3.3.** *If the Eisenstein datum starts from a trivial representation, then for $\mathrm{Re}(s) \gg 0$ for each complex variable, $\mathcal{O}_G(E(\pi, \overline{s})) = \{\mathbf{a}\}$, where $\mathbf{a}$ is the transpose of the partition corresponding to the parabolic subgroup we are inducing from.*

**Proposition 3.1.** (1) *The above conjecture is true for a maximal parabolic.*

(2) *The generalized Speh representation $E_\tau(g)$ has $\mathcal{O}_G(E_\tau(g)) = (m^r)$, where $\tau$ is a generic representation on $\mathrm{GL}_m(\mathbb{A})$, $E_\tau(g) = \mathrm{Res}_{\overline{s}=\overline{s}_0} E_\tau(g, \overline{s})$, $E_\tau(g, \overline{s})$ is the Eisenstein series corresponding to $\mathrm{Ind}_{P_{m,m,\cdots,m}(\mathbb{A})}^{\mathrm{GL}_{rm}(\mathbb{A})}(\tau|-|^{s_1} \otimes \cdots \otimes \tau|-|^{s_r})\delta_{P_{m,\cdots,m}}^{1/2}$ and $\overline{s}_0 = ((r-1)/2, (r-3)/2, \cdots, -(r-3)/2, -(r-1)/2)$.*

3.2. **General automorphic forms.**

**Conjecture 3.4.** *For any special unipotent orbit $\mathbf{c}$ of $G$, there is an automorphic representation $\pi$ such that $\mathcal{O}_G(\pi) = \{\mathbf{c}\}$.*

**Conjecture 3.5.** *For any automorphic representation $\pi$ of $G(\mathbb{A})$, every unipotent orbit of $\mathcal{O}_G(\pi)$ has the same dimension.*

## 4. Residual spectrum

For $\mathrm{GL}_n$, residual spectrum is completely understood by Moeglin-Waldspurger. Basically they are all given by Speh representations:

**Definition 4.1.** *For $n = ab$, let $\tau$ be a cuspidal unitary autrep of $\mathrm{GL}_a$. Then Eisenstein series built out of $P_{a^b}$ and $\tau^{\otimes b}$ has $b$ complex variables, and has a pole at $s = (\frac{b-1}{2}, \frac{b-3}{2}, \cdots, \frac{1-b}{2})$ so that*

$$E_{-1}(\phi, g) = \lim_{s \longrightarrow \textit{that point}} \prod_i (s_i - s_{i+1} - 1) E(\phi, s, g),$$

*is an $L^2$ automorphic form on $\mathrm{GL}_n$. Denoted as $\mathcal{E}_{(\tau,b)}$, these are exactly those appearing in the residual spectrum of $\mathrm{GL}_n$.*

## References


[ColMcG] D. Collingwood, W. McGovern, *Nilpotent orbits in semisimple Lie algebras.* '91.

[Ginzburg] D. Ginzburg, *Certain conjectures relating unipotent orbits to automorphic representations.* Israel J. Math. '06.